Sebastian Alonso Gonzales-Portillo

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Education

University of Miami – Dual Degree **Coral Gables, FL**

B.S. in Computer Science & Mathematics; B.S.B.A. in Business Technology, cum laude *May 2025*

- **Minor**: Astrophysics **GPA**: 3.84/4.0
- **Relevant Coursework**: Data Structures & Algorithm Analysis, Operating Systems, Computer Networks, Database Management Systems, Software Engineering, Machine Learning, Numerical Analysis, Numerical Linear Algebra

Technical Skills

Languages: Python, SQL, Java, C
Backend/Architecture: FastAPI, Pydantic, SQLAlchemy, REST APIs, asynchronous programming, microservices
Cloud/DevOps: GCP, Cloud Run, Cloud Functions, GKE/Kubernetes, Docker, GitHub Actions, Cloud Build, Secret Manager, CI/CD
Data/Workflows: PostgreSQL, PostGIS, BigQuery, Bigtable, Firestore, Redis, ETL, batch processing, job orchestration
ML/Operations: PyTorch, scikit-learn, XGBoost, LightGBM, CatBoost, FAISS, Sentence Transformers, OpenAI API, Pytest, Grafana, Google Cloud Logging

Experience

PadStats **Hybrid – Miami, FL**

Founding Software Engineer – Cloud & ML Systems *June 2025 – June 2026*

Software Engineering Intern *June 2024 – May 2025*

- Architected and built a production property-intelligence platform on GCP using Python, FastAPI, Cloud Run, PostgreSQL/PostGIS, and BigQuery, Redis, supporting valuation models, geospatial analytics, and customer-facing property-intelligence products.
- Designed and trained a multi-task computer-vision model on 15K+ labeled real-estate images using an ImageNet-pretrained backbone, predicting room type from a company-defined taxonomy, Fannie Mae-aligned construction-quality attributes, furnishing status, and other property features used across company products.
- Built containerized GPU-backed training, evaluation, and batch-inference workflows on GKE for concurrent computer-vision experiments, alongside a Streamlit/Firestore labeling and QA platform supporting iterative dataset refinement, atomic task locking, review queues, and model-version evaluation.
- Deployed the vision system behind a FastAPI/PyTorch inference service, increasing batch throughput 10–12x through asynchronous image retrieval, bounded concurrency, optimized preprocessing, and batched GPU inference.
- Designed a distributed model-serving architecture in which a FastAPI orchestration service concurrently invoked independently deployed XGBoost, LightGBM, and CatBoost models, with registry-driven model selection and versioning across U.S. regions, TTL caching, prediction aggregation, and failure isolation.
- Developed an asynchronous Python ETL system ingesting crime data from 25 U.S. cities across ArcGIS, Socrata, CKAN, CSV, Excel, and scraped sources, with Pydantic validation, idempotent upserts, deduplication, Alembic migrations, and run-state tracking.
- Engineered an embedding-based NLP classification system that mapped noisy crime descriptions to FBI NIBRS codes using Sentence Transformers, FAISS, nearest-neighbor voting, and confidence-based human-review thresholds.
- Partnered directly with company leadership to translate ambiguous property-risk and location-intelligence requirements into 11+ customer-facing property-intelligence APIs spanning natural hazards, crime, census and socioeconomic data, comparable properties, and bundled intelligence reports.
- Instrumented production APIs, ETL pipelines, and ML workflows with SQL monitoring views, Grafana dashboards, freshness and error metrics, Google Cloud Logging, and unit, integration, and dry-run end-to-end testing for safer refactors and production cutovers.
- Standardized containerization and CI/CD across APIs, ETL jobs, and ML services using Docker, GitHub Actions, and Cloud Build, supporting both Cloud Run services and GPU or long-running workloads on GKE.

Bonsai Applied Computations Group **Coral Gables, FL**

Project Manager / Analyst *January 2024 – May 2025*

- Led a team of analysts processing 650,000+ social-media posts with OpenAI API-assisted sentiment analysis and time-series visualizations related to the 2024 U.S. presidential election.
- Led modernization work for a MariaDB-backed astrophysical catalog, improving query performance and data-access patterns across a dataset containing more than 7 million records.

University of Miami College of Arts & Sciences **Coral Gables, FL**

Machine Learning Research Assistant *September 2023 – May 2025*

- Built PyTorch experiment workflows for training and evaluating ResNet and DenseNet models across MNIST, FashionMNIST, CIFAR-10, and CIFAR-100 to analyze learned representations, training dynamics, and neural collapse.